Agri Rover

A cable-suspended rover that monitors crops from above

Fokir Mostakim Billa Sobuj
0112430197

Shaikh Jubair
0112420017

Tahmidur Rahman Foysal
0112430199

Najif Jawoad Jim
0112330545

Md. Rubyat Simum Mahi
0112330575

Jakaria Shikder
112230715

The problem with field monitoring

Slow manual checks

Walking every row takes hours and varies by person.

Late detection

Disease, pests and weeds spread before anyone notices.

Rovers get stuck

Wheeled robots sink in mud and trample crops.

No live data

Farmers lack continuous readings of field conditions.

A rover that hangs above the crops

No ground contact

A rope and pulley ropeway means no crop damage and no mud.

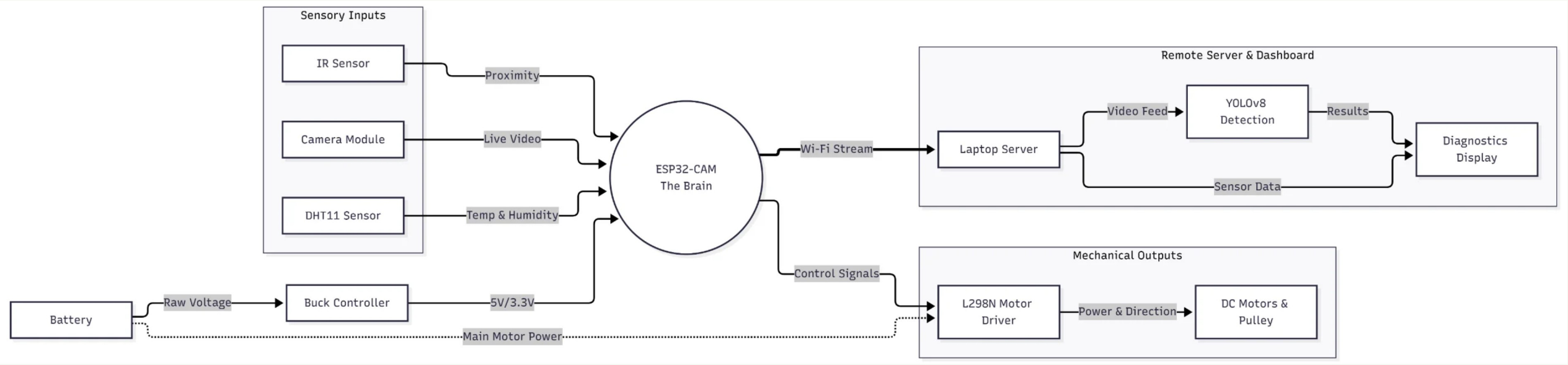
Autonomous

Moves, stops, scans and reverses on its own.

AI and live data

YOLOv8 diagnosis and sensor readings on one dashboard.

System architecture: data flow map



Hardware components

Component	Role
ESP32-CAM	Core controller and live video
L298N motor driver	Motor speed and direction
DC motors and pulley	Drive the rover along the rope
DHT11	Temperature and humidity
IR sensor	Detects poles and rope end for auto-reverse

The stop-and-scan cycle

2 s

Move along the rope

5 s

Stop, scan crops, read DHT11

Reverse

IR sensor spots the pole and flips direction

The cycle repeats back and forth across the whole field.

YOLOv8 crop diagnostics

Video frames from the ESP32-CAM stream to a Python backend, where YOLOv8 marks what it finds and sends annotated images to the dashboard.

Diseases

Leaf and plant infections

Pests

Insects on leaves and stems

Weeds

Unwanted plants between rows

Real-time dashboard

Live video

See what the rover sees as it scans.

Temperature and humidity

Current DHT11 readings at each stop.

AI diagnostic images

Detections marked with boxes and labels.

Detection history

Track crop health over time.

Impact and next steps

Advantages

- No crop damage
- Works in mud
- Low cost
- Early detection

Challenges

- Limited to rope path
- Wind and weather
- Wi-Fi range
- DHT11 accuracy

Future scope

- Solar power
- Precision spraying
- Mobile alerts
- Cloud analytics

Thank you

Smarter fields, healthier crops, better yields.

Questions?